GILDED CIRCUIT · ลายรดน้ำ × ลายวงจร

Kanokpon Mettasat

วิศวกรไฟฟ้าและคอมพิวเตอร์ · ลายกนกไทยที่กลายเป็นลายวงจร

Hello — I'm Gun. A Thai army cadet on a five-year scholarship in the US, studying engineering with honors, and I love building software people can actually use. Everything on this site is real and running.

> Electrical & Computer Engineering, Institute Honors Program -- post-quantum cryptography & blockchain security research · Virginia Military Institute, Class of 2027 · Overall GPA 3.81 / Major GPA 4.00 -- Institute Honors Program

> The mark is literal: กนก (kanok) — the first word of the name — means gold, and names the Thai flame ornament itself.

EXPLORE สำรวจ

About
The short story — cadet, engineer, builder.

Education
4 schools, two countries — top of the class.

Projects
Research, hardware, shipped software — see what they do.

Audience-specific: recruiters · interns · grad school

More: notes · reading list · null registry

BUILD ACTIVITY recent commits

CONTACT ติดต่อ

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GitHub · LinkedIn · ORCID

ABOUT

เกี่ยวกับ

Kanokpon Mettasat

Electrical & Computer Engineering, Institute Honors Program -- post-quantum cryptography & blockchain security research · Virginia Military Institute, Class of 2027 · Overall GPA 3.81 / Major GPA 4.00 -- Institute Honors Program

Electrical and Computer Engineering cadet at **Virginia Military Institute** (Class of 2027, Institute Honors Program, major GPA 4.00), applying to MS programs in Electrical Engineering for Fall 2027 with a focus on **secure and embedded systems and post-quantum cryptographic engineering** for resource-constrained devices. Research: undergraduate researcher with Dr. Efat Fathalla (VMI ECE) on quantum-resistant blockchain security -- first-author paper accepted at **IEEE UEMCON 2026**, a second published in Computers (MDPI), and a third manuscript in preparation from the Institute Honors thesis. Engineering: hardware lead on a KiCad PCB from schematic to fabricated board; team lead at IEEE SoutheastCon 2026 Undergraduate Circuit Design (top seven of forty teams); sole developer of **Officer Parker**, a real-time voice agent for mock US visa interviews; senior design capstone on a wearable sensor sock. Independent work: a Python statistical research platform for financial time series, bound to acceptance thresholds registered before any experiment ran. Service: President of Eta Kappa Nu (IEEE-HKN) at VMI for 2026-27; math tutor in the VMI Open Mathematics Laboratory since 2024. Before VMI: Chulachomklao Royal Military Academy (Class 74) and the Armed Forces Academies Preparatory School, Thailand, where I ranked first of 348 Army cadets; selected first of the class for the Royal Thai Army's overseas study programme to the United States.

PUBLICATIONS

ผลงานตีพิมพ์

Oct 2026

K. Mettasat and E. Fathalla. “Post-Quantum Cryptography for Blockchain: A Systematic Mapping by Cryptographic Family.” *2026 IEEE 17th Annual Ubiquitous Computing, Electronics & Mobile Communication Conference (UEMCON), Armonk, New York.*

Accepted; to appear in IEEE Xplore

Oct 2026

K. Mettasat, E. Fathalla and M. Azab. “A Systematic Mapping of Quantum Threats and Defenses for Blockchain.” *Computers (MDPI), vol. 15, no. 11, art. 743.* doi:10.3390/computers15110743

Published; open access

2027

K. Mettasat and E. Fathalla. “Post-quantum-secure blockchain architecture (manuscript arising from the Institute Honors thesis).”

In preparation

PAPERS, TALKS AND FIGURES →

ORCID: orcid.org/0009-0002-6260-8196

SKILLS

ทักษะ

LANGUAGES

Python | C | C++ | MATLAB | JavaScript / TypeScript | PowerShell | Bash

HARDWARE

KiCad (schematic to fabricated PCB) | LTspice / Multisim | MATLAB / Simulink | Arduino and Raspberry Pi | I2C / SPI / UART | Oscilloscope, soldering and board assembly

ENGINEERING

Digital systems design | Microcontrollers | Circuit design and analysis | Signal processing

RESEARCH

Systematic review methodology | Statistical hypothesis testing | Bootstrapped / deflated Sharpe ratio | IEEE technical writing

TOOLING

Git | pytest | pandas / NumPy | pyright | Playwright

WEB / INFRASTRUCTURE

FastAPI | Jinja2 | Linux and Windows server administration | Scheduled-job automation | SQLite

LIFE AT VMI

ชีวิตนักเรียนนายร้อย

OFF DUTY

นอกเวลางาน

EDUCATION

การศึกษา

From a Thai military academy to an honors engineering program in Virginia -- 4 schools, two countries, one through-line: finish first, then go further.

AUG 2023 -- MAY 2027 (EXPECTED)

Virginia Military Institute

BS, Electrical and Computer Engineering · *Lexington, VA*

- ✦ **Institute Honors Program, VMI (selected Jan 2024) -- Honors scholarship since Jul 2024**
  - Dean's List, multiple terms
- ✦ **Overall GPA 3.81 / Major GPA 4.00**
  - Term GPA 4.00 (Year 2 Term 1)
  - Term GPA 4.00 (Year 2 Term 2, all-A semester)
  - Term GPA 3.81 (Year 3 Term 1)
  - Term GPA 3.70 (Year 3 Term 2)
  - President, Eta Kappa Nu (IEEE-HKN), VMI chapter (2026 -- 2027)
- ✦ **Tau Beta Pi -- Engineering Honor Society, Virginia Chapter (inducted Nov 2025)**
- ✦ **IEEE SoutheastCon 2026 -- top 7 of 40 teams, Undergraduate Circuit Design (Alabama, Mar 2026)**
- ✦ **Tau Beta Pi Convention -- VMI representative (Mar 2026)**
  - OGA Representative, Hotel Company -- oversees new-cadet orientation (May 2026 -- present)

- Cadre Old Corps Corporal + Cadre Rat Challenge (2024 -- 2025); MathLab tutor, VMI library (2024 -- present)
- Expert classification, Rifle (May 2025); 9 mm pistol qualification (2025)
- Commendation from the Chief of Defence Forces, Royal Thai Armed Forces (Apr 2024)
- Community service in Puerto Rico with the VMI Glee Club (Mar 2024)
- Coursework: Signal & System Analysis, Computer Architecture, Digital Systems Design, Microcontrollers, Electronics, Probability & Statistics for Engineers, Discrete Mathematics, Differential Equations; Senior Design Capstone (EE-421X, in progress)

AUG 2022 -- MAY 2023

University of Colorado Boulder

International English Center, Levels 4-7 · *Boulder, CO*

- Certificate of completion, English Programs

MAR 2022 -- AUG 2022 (PRE-US DISPATCH)

Chulachomklao Royal Military Academy (CRMA) -- Class 74

Army cadet, English Program (graduating class) · *Thailand*

- Selected first of the class for the Royal Thai Army's overseas study programme to the United States (May 2022)
- Section Leader, Section A.1 (EP)

JUL 2020 -- FEB 2022

Armed Forces Academies Preparatory School (AFAPS) -- Class 63

Pre-cadet, Potential English Program (PEP) · *Thailand*

- Ranked first of 348 Army cadets across multiple semesters (cumulative GPA 3.96-3.97)
- Ranked #3-4 of 878 in the overall preparation class
- Vice-President, Army cadet section + Honor Council insignia (Jul 2021)
- Platoon Leader insignia, 2nd Squad / 2nd Battalion (Jun 2021)
- Honorable Mention, 20th National Physics Olympiad (Dec 2021)
- Outstanding Youth in Physics -- Science Association of Thailand under Royal Patronage (Dec 2021)
- ALCPT silver medal + AFAPS completion certificate from the Commander-in-Chief (Feb 2022)
- Seven academic certificates of appreciation (Feb 2022)

Honors, awards, leadership (13) 2021-07 -- 2026-09

2026-09

President, Eta Kappa Nu (IEEE-HKN), VMI chapter, 2026-27

@ VMI

2026-05

OGA Representative, Hotel Company -- new-cadet orientation lead

@ VMI

2026-03

IEEE SoutheastCon 2026 -- Top 7 of 40 teams, Undergraduate Circuit Design

@ Alabama

2026-03

Tau Beta Pi Conference -- VMI representative

@ Tau Beta Pi Society

2025-11

Tau Beta Pi -- Engineering Honor Society, Virginia Chapter

@ VMI

2025-05

Expert classification, Rifle

@ VMI

2024-07

VMI Institute Honors Program scholarship (USD five thousand / year)

@ VMI

2024-04

Commendation from the Chief of Defence Forces

@ Royal Thai Armed Forces

2024-01

Selected into the VMI Institute Honors Program

@ VMI

2022-05

Selected first of the class for the Royal Thai Army's overseas study programme to the United States

@ Chulachomklao Royal Military Academy (CRMA)

2022-02

AFAPS completion + ALCPT silver medal (Commander-in-Chief)

@ AFAPS

2021-12

Honorable Mention, 20th National Physics Olympiad

@ Thailand

2021-07

Vice-President, Army cadet section + Honor Council

@ AFAPS

LATEST TERM • NEW

อัปเดตเทอมล่าสุด

2024-2026 update

Year 2 (VMI, Aug 2024 -- May 2025). Coursework in the ECE core: Digital Systems Design, Electrical Circuit Analysis, Discrete Mathematics, multivariable Calculus. Closed Term 2 with a Term GPA of 4.00 (all-A semester).

Summer 2025 (Bangkok). Returned home; spent the off-term on independent quantitative-trading research (pre-registered, walk-forward validated).

Year 3 Term 1 (VMI, Aug 2025 -- Dec 2025). Computer Architecture, Microcontrollers, Electronics, Modern Physics. Term GPA 3.81. **Tau Beta Pi** (Engineering Honor Society, Virginia Chapter) invitation in Nov 2025.

Year 3 Term 2 / Spring 2026 in progress. Honors-program coursework; competed at **IEEE SoutheastCon 2026** in Alabama and placed **4th in Undergraduate Circuit Design** (Mar 2026). Represented VMI at the **Tau Beta Pi national conference** the same month.

SOFTWARE SHIPPING CADENCE (RESEARCH-ONLY, PUBLIC-REPO DATES)

- **May 2026 -- LINE OA personal-secretary bot.** Multi-owner LINE Messaging API service with daily 18:00 BKK eve-of briefing, 1-hour pre-alerts, Sunday weekly digest, daily SQLite snapshot backup, Telegram-style task commands (`/list` , `/done` , `/retry`). Deployed unattended on a Windows VPS behind Tailscale. ~201 tests.
- **May 2026 -- Research dashboard.** `FastAPI` + Jinja2 monitoring site with multiple pages: hub with a BKK/NY clock tower, project board with phase-locked Gantt rows, 30-agent organisation chart, pixel-art floor-plan view of active agents, and a decision-rules anchor page. Hard-restart deploy via PowerShell; reachable on a Tailscale Funnel URL. ~300+ tests.
- **May 2026 -- /portfolio public personal-introduction page.** This page. `FastAPI` + Jinja2 with an IDE-shell layout, multi-route split, view-source modal, Cmd-K command palette, keyboard-driven `vim` -style navigation, and a 365-day commit-activity grid. ~120 tests on this surface alone.

A companion systematic mapping followed a **PRISMA 2020 protocol** across **387 peer-reviewed studies**, published in *Computers* (MDPI) in 2026.

PROJECTS

ผลงาน

Research, hardware and shipped software — built around cadet life. Every card opens in plain words first (what it is, why it matters), then goes technical for the engineers. อ่านง่ายก่อน แล้วค่อยลงลึกแบบวิศวกร

Officer Parker -- real-time voice agent for mock US visa interviews

- Practise a US visa interview out loud with an AI consular officer who already has your own form on screen -- and catches every place your answers stop matching it.

Browser voice agent that runs a two-minute mock US visa interview end to end: 24 kHz PCM capture, a speech-to-speech WebSocket session, and a structured report generated from the transcript. Grounded in the applicant's own DS-160 answers so the agent can surface contradictions between what was filed and what is said -- a check a generic mock-interview tool cannot make. Sole author of the architecture, front end, serverless functions, deployment and submission video. Submitted to the AssemblyAI Voice Agent Hackathon (Sep 2026); MIT-licensed and running in production.

JavaScript | Web Audio (AudioWorklet, 24 kHz PCM) | WebSocket speech-to-speech | Vercel serverless functions | AssemblyAI

// Sole developer, Aug -- Sep 2026; zero runtime dependencies; guard tests for the honesty constraint and the client-side rate limiter, each verified against a mutation battery

LIVE DEMO ↗ · SOURCE ON GITHUB (MIT) ↗

READ DEEP-DIVE →

Statistical research platform for financial time series

- A home-built science lab for testing ideas about market data: every hypothesis has to clear statistical thresholds that were fixed before the experiment ran.

Independent project, 2025 -- present. A Python research stack -- data pipeline, event-driven simulator with realistic transaction costs, telemetry -- bound to statistical acceptance thresholds registered before any experiment ran (bootstrapped lower-bound Sharpe ratio, deflated Sharpe ratio, multiple-testing correction). Over sixty hypotheses tested and rejected by the data rather than by discretion; automated audits for look-ahead and optimistic fills, and mutation-tested guards on the execution path.

Python | pandas / NumPy | Event-driven simulator | Block bootstrap | pytest + mutation testing

// 60+ hypotheses tested; companion systematic mapping (SURI) on the literature side

READ DEEP-DIVE →

SURI -- Systematic mapping of quantum-resistant blockchain security

- I mapped what the research world knows about quantum computers threatening blockchains -- following a protocol written down before the search began, so anyone can rebuild the same map.

Undergraduate research with Dr. Efat Fathalla (VMI ECE), 2025 -- present. A systematic mapping of quantum-resistant blockchain security: 387 peer-reviewed primary studies (2018-2026) selected from more than 6,500 records under a PRISMA 2020 protocol, then classified by contribution type and evidence maturity. The review pipeline is implemented in Python as idempotent build / append / repair / categorise stages, so the whole corpus can be rebuilt from the search log after any protocol amendment. A mid-study reclassification was reconciled by correcting the earlier categorisation rather than defending it, and documented in the protocol record. Output so far: a first-author paper accepted at IEEE UEMCON 2026 (Armonk, NY) and a second paper published open access in Computers (MDPI).

IEEE Xplore + MDPI | PRISMA 2020 | Evidence-maturity scale (L0-L5) | Multi-label coding | Python pipeline

// 387 included studies from 6,500+ records; 2 research questions; 6-level evidence-maturity scale; two papers, one published and one accepted

PUBLISHED PAPER ↗ · DOI ↗

READ DEEP-DIVE →

IEEE SoutheastCon 2026 -- Undergraduate Circuit Design (team lead)

- A circuit-design competition where the problem is handed out that morning and the board has to work by the afternoon.

Team lead and principal designer. Designed, built and tested a digital circuit -- combinational logic with a microcontroller -- on site from a problem released that morning, under a half-day limit, leading a team of four. Top seven of forty teams from universities across the southeastern United States (Alabama, Mar 2026).

Combinational logic | Microcontroller | Breadboard prototyping | On-site debugging

// Top seven of forty teams; Mar 2026

Senior Design Capstone (EE-421X) -- wearable sensor sock

- A year-long team project to design and build a sock with sensors in it, run the way a real engineering programme is run: requirements first, then design, then build.

Team member, Aug 2026 -- May 2027. Team project run as a formal systems-engineering process; contributed to the System Requirements Document and the September requirements review (concept of operations, functional block diagram, threshold and objective requirements); design and build follow.

Systems-engineering process | System Requirements Document | Concept of operations | Functional block diagram

// Aug 2026 -- May 2027; requirements review Sep 2026; design and build in progress

Motion-triggered novelty device (EE-382) -- schematic to fabricated PCB

- A small gadget that senses light, sound and motion and plays audio back -- I designed the circuit board from scratch, had it manufactured and soldered it by hand.

Hardware lead, EE-382 Entrepreneurship for ECE, Spring 2026. Sole designer of the electronics: schematic and printed circuit board in KiCad, sent to fabrication, hand-assembled, and firmware for an Arduino-class microcontroller fusing light, sound and motion sensing to trigger audio playback. Debugged sensor and audio paths on the oscilloscope; teammates carried customer discovery and the business case.

KiCad (schematic + PCB) | Arduino-class microcontroller | Light / sound / motion sensors | Audio playback | Oscilloscope debugging

// Spring 2026; sole designer of the electronics; board fabricated and hand-assembled

EARLIER WORK

ผลงานก่อนหน้า

Software systems from 2025 -- 2026, still running. Kept for the engineering record; the research and hardware work above is current.

Trading-research dashboard

- A live mission-control screen for my research projects -- one glance from any device shows what every system is doing right now.

FastAPI + Jinja2 monitoring and decision-support dashboard for a personal quantitative-trading workflow. Surfaces include project tracking, an editorial organisation chart, a Phaser 3 isometric office-RPG view of active research agents, a live-demo strategy monitor with a tail-concentration auditor, and a Smart Money Radar that scores 13F / Form-4 / political-disclosure signals (1-100). Deployed via Tailscale Funnel to a Windows VPS with a hard-restart deploy script.

Python | FastAPI | Jinja2 | Phaser 3 | pytest | PowerShell

// ~679 tests, 12+ public/private pages, zero-downtime redeploy

READ DEEP-DIVE →

LINE Official Account bot

- A chat assistant that never forgets: it lives in the LINE app, remembers everyone's plans, and sends the reminders all by itself.

Multi-owner LINE Messaging API bot with scheduled reminders, daily SQLite snapshots, Sunday digest, and pre-alert dedupe. Designed to run unattended on a Windows VPS with a self-healing restart loop.

Python | FastAPI | SQLite | LINE Messaging API

// ~201 tests, multi-tenant

READ DEEP-DIVE →

Money Compass -- personal wealth aggregator

- A private app that reads my own bank statements and shows where the money actually goes -- like a personal accountant who never sleeps.

A read-only personal-finance aggregator that pulls Bank of America and US credit-card transactions through the SimpleFIN Bridge and turns them into a cashflow, runway, and subscription view. It is a package inside the trading dashboard, exposed at an auth-gated /money route that fails closed and never renders financial figures on a public surface. A daily refresh task bakes the account state into a JSON snapshot so the request path never touches the bank API or the SQLite ledger directly. Phase 1 (ledger + cashflow + subscription detect) and Phase 2 (month-over-month, budget streak, goals, ranked CFO insights, spending-trend chart, savings tips, category drill-down) both ship and run live on the VPS.

Python | FastAPI | SQLite | SimpleFIN Bridge | Jinja2 | uPlot | pytest

// 40+ tests, Phase 1 + Phase 2 live, auth-gated + leak-guarded

READ DEEP-DIVE →

Edge Factory -- research-automation pipeline

- An assembly line for research ideas: drop an idea in one end and it gets tested the same fair way every time -- most ideas die cheap, and the rare good one earns a full report.

A thin Python orchestrator that runs a strategy candidate down a nine-station conveyor -- seed, expand, dedup, prescreen, backtest, bias-audit, statistical gate, dossier, human verdict -- so systematic-trading ideas are evaluated the same way every time. The design point is not generating many ideas but killing bad ones cheaply and early: a factory without a hard kill-gate just produces false positives at high speed. Two safety invariants are locked in code -- the pipeline always stops at the dossier and never has a path to live capital, and no agent can approve its own candidate. The MVP wires the dedup, prescreen, and statistical gate automatically around a reused fleet of existing research tools.

Python | pandas | pytest | pyright

// MVP v0.1 committed, 36 pytest passing, pyright 0, forbidden-import guard 0

READ DEEP-DIVE →

Live web karaoke -- group mode

- Turns your phone into a karaoke microphone for the TV -- nothing to install, just scan a code and sing.

A no-install progressive web app that turns a phone into a karaoke microphone and streams the vocal to a TV or laptop over WebRTC, with a QR-code join flow and a group party mode as the core use case. The hard engineering problem was audio latency: the singer hears their own voice through the air instantly and again from the speaker late, so the app drives the receiver's jitter buffer down to remove the browser's default 100-200ms of playout padding. It is deployed live on Vercel and stays within YouTube's terms by embedding the official iframe player rather than scraping audio. A service worker caches the shell for offline-capable reloads, with an explicit cache-version discipline after a stale-render incident.

JavaScript (ES modules) | WebRTC | PeerJS | YouTube IFrame API | PWA / Service Worker | Vercel

// Live on Vercel, v0.5.1, group-mode validation in progress

READ DEEP-DIVE →

Trading Journal -- decision-intelligence layer

- A diary that grades my own decisions: when I overrode the system by hand, did that actually help? The diary answers with numbers, not feelings.

A read-only decision-intelligence layer over a real-money trading track that scores whether the operator's manual closes actually beat the bot's own trailing exit, then narrates the day as a short desk note. It never touches the edge or any order path -- it only reads the realised trade log and the divergence report. The daily note is written by a Groq-hosted model as decorative prose while the numbers stay authoritative, and it is exposed at an auth-gated /journal route rendered as a desk-blotter page with a P&L calendar heatmap. Phase 1 (the deterministic scorecard) is live on the VPS with an hourly producer task.

Python | FastAPI | Groq API | Jinja2 / server-rendered HTML | Telegram | pytest

// Phase 1 live on VPS, 15 tests, auth-gated + metric-drift tripwire

READ DEEP-DIVE →

CONTACT

ติดต่อ

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One-page CV: download PDF

Last updated: 2026-10-07. Content here is a public summary; the formal CV is available on request.